

Colin Zeng

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EDUCATION

Duke University

GPA: 3.93/4.0

Bachelor of Science in Computer Science and Statistics

Expected May 2028

- Relevant Coursework: Machine Learning, Computer Systems, Database Systems, Data Structures and Algorithms, Multivariable Calculus, Linear Algebra, Advanced Probability, Regression Analysis
- Extracurriculars: Catalyst (Duke's Premier Tech Society), Machine Learning, Statistical Science Majors Union
- Awards: Ultimate Frisbee U20 World Champion, Team USA, World Junior Ultimate Championships

EXPERIENCE

Undergraduate Research Intern

Jan. 2026 – Present

Dietrich Lab @ Duke University | fungalpangenome.ccn.duke.edu

Durham, NC

- Designed a PostgreSQL schema mapping 1,482 yeast strains across core genes and pangenome traits
- Built a full stack frontend (React, JavaScript, HTML/CSS) with SQL backed queries loading 10,000+ genes across *Saccharomyces cerevisiae* and *Ashbya gossypii* pangenomes
- Utilized AlphaFold and BLAST across 6,294 genes for protein prediction and sequence alignment in the pipeline

Software Engineer Intern

Oct. 2024 - Aug. 2025

Quantworks/SIZEO

Durham, NC

- Engineered a web platform using Anthropic's Claude API to optimize budget allocation, packaging, and seasonal inventory predictions for retail clients including JCPenney and Aeropostale
- Built Python pipelines turning client uploaded CSVs into multi-trend sales and demand views in Streamlit and Plotly
- Designed and presented 30 client dashboard screens in Figma, introducing an artifact gallery of saved LLM graphs and reports for downstream RAG analysis, and for planners to revisit and compare

Research Intern

June 2023 - July 2024

NCSSM Summer Research Programs

Durham & Wilmington, NC

- Programmed an agent based Hodgkin Huxley neuron simulation in NetLogo (4 ion species, 6 channel types)
- Modeled a cholera transmission system extended with perfect and imperfect vaccinations (5 coupled ODEs)
- Modeled real gas behavior in Gaussian; presented all 3 projects at Summer Ventures and SRIP

PROJECTS

Speech Dysfluency Detection and Removal | *Python, Whisper, scikit-learn*

- Built an end-to-end pipeline to detect, categorize, and remove speech dysfluencies (e.g., repetitions)
- Used OpenAI Whisper to transcribe 162 audio clips and rule based logic to flag and categorize 5 dysfluency types (repetition, insertion, deletion, pause, substitution)
- Benchmarked 3 classifiers on an 88 sample test set; a NeMo model reached 79.6% accuracy (0.78 macro F1), a 44% gain over an MFCC random forest baseline and 3 times as accurate as the Llama 3 baseline

Insight, Journal Reflection App | *React, Groq API, LLM Integration*

- Built a React app that generates personalized reflective prompts from journal entries via the Groq API
- Designed retrieval system from scratch using vectorization and cosine similarity to query relevant entries for RAG
- Shipped a new user onboarding flow, 5 selectable themes, and persistent entry management backed by localStorage

Ultimate Frisbee Play Classifier | *PyTorch, YOLOv8, Computer Vision*

- Built a 2 stage computer vision pipeline that classifies 7 ultimate frisbee plays from images by detecting players and the disc, then converted detections into simplified dot-map representations to train a downstream play classifier
- Fine tuned YOLOv8 on a Roboflow dataset, reaching 85% precision and 87% recall across 3 object classes
- Benchmarked ResNet18 for classification, reaching 91.3% accuracy and outperforming custom CNN by 20 points

Event Horizon, Unreal Engine 5 Game | *C++, Unreal Engine 5, Blueprints*

- Built a 3D precision platformer in Unreal Engine 5 (kooling.itch.io), using Blueprints for in-game triggers and UI
- Wrote reusable C++ components with adjustable features, adding sliding and bouncing behavior to platforms

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript, React, R, C++, HTML/CSS

Frameworks & Libraries: React, PyTorch, YOLOv8, scikit-learn, pandas, NumPy

APIs & AI Integration: Groq API, Anthropic Claude API, Whisper, LLM Prompt Engineering

Developer Tools: Git, Unix/Linux, Figma, Roboflow, LaTeX, Google Colab, Plotly, Streamlit